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Project:	Matrix Application in Cryptology

Objectives:

- (1.) Encode a secret message using a 2×2 matrix.
- (2.) Decode the secret message encrypted/encoded with a 2×2 matrix.
- (3.) Encode a secret message using a 3×3 matrix.
- (4.) Decode the secret message encrypted/encoded with a 3×3 matrix.

Cipher Key:

A	B	C	D	E	F
1	2	3	4	5	6
G	H	I	J	K	L
7	8	9	10	11	12
M	N	O	P	Q	R
13	14	15	16	17	18
S	T	U	V	W	X
19	20	21	22	23	24
Y	Z		?	!	.
25	26	27	28	29	30

(1.) Scenario/Story for the 2 by 2 encoded secret message:

Given: You are playing a game of hide and seek with your childhood friends, Alex and Brian. You and Brian are sent to go hiding while Alex counts down

from 10 to start seeking. Before the game started, you developed the above Cipher Key to communicate. Each letter and character directly correspond to a certain number. This allows the both of you to communicate without making noise or leaving messages that Alex can understand. You both taped a piece of paper on the door to each room of the house to develop a strategy. Anytime one of you saw or heard Alex enter the room, search for players, and leave when he couldn't find anyone, you would write the following message:

ALEX WAS HERE!

With the following decoded message being

1 12 5 24 27 23 1 19 27 8 5 18 5 29

Using matrices, we encode the secret message using a 2×2 encoding matrix. To start, a decoded message matrix with 2 rows is required to deal with the dimensions of the encoding matrix. Using the decoded message above, the matrix is:

$$\begin{bmatrix} 1 & 5 & 27 & 1 & 27 & 5 & 5 \\ 12 & 24 & 23 & 19 & 8 & 18 & 29 \end{bmatrix}$$

After the setup of the decoded message matrix above, an invertible encoding matrix with only positive integers must be found. The inverse of that matrix needs to be the decoding matrix (different from the decoded message matrix above) and must also have integers for all the elements in the matrix. The encoding matrix used is:

$$A = \begin{bmatrix} 3 & 1 \\ 8 & 3 \end{bmatrix}$$

To find the encoded message (B), the encoding matrix must be multiplied by the decoded message matrix. To multiply, the number of columns in the encoding matrix must equal the number of rows in the decoded message matrix. Since both

of these values are equal to 2, the matrices can be multiplied and the encoded message can be found following the steps below:

$$B = \text{Encoded Message} = \begin{bmatrix} 3 & 1 \\ 8 & 3 \end{bmatrix} * \begin{bmatrix} 1 & 5 & 27 & 1 & 27 & 5 & 5 \\ 12 & 24 & 23 & 19 & 8 & 18 & 29 \end{bmatrix}$$

First row calculations:

$$3(1) + 1(12) = 15$$

$$3(5) + 1(24) = 39$$

$$3(27) + 1(23) = 104$$

$$3(1) + 1(19) = 22$$

$$3(27) + 1(8) = 89$$

$$3(5) + 1(18) = 33$$

$$3(5) + 1(29) = 44$$

Second row calculations:

$$8(1) + 3(12) = 44$$

$$8(5) + 3(24) = 112$$

$$8(27) + 3(23) = 285$$

$$8(1) + 3(19) = 65$$

$$8(27) + 3(8) = 240$$

$$8(5) + 3(18) = 94$$

$$8(5) + 3(29) = 127$$

Therefore, the encoded message is

$$B = \begin{bmatrix} 15 & 39 & 104 & 22 & 89 & 33 & 44 \\ 44 & 112 & 285 & 65 & 240 & 94 & 127 \end{bmatrix}$$

Or written out as

$$15 \ 44 \ 39 \ 112 \ 104 \ 285 \ 22 \ 65 \ 89 \ 240 \ 33 \ 94 \ 44 \ 127$$

(2.) Decoding the secret message encoded with 2 by 2 matrix:

The second part of this 2 * 2 encoded matrix is to decode the secret message, which we can now do. You saw Alex leave one of the bedrooms earlier from your hiding spot, and you decided to quickly write on the piece of paper:

$$15 \ 44 \ 39 \ 112 \ 104 \ 285 \ 22 \ 65 \ 89 \ 240 \ 33 \ 94 \ 44 \ 127$$

Alex will likely take a while before double-checking a previously visited room, so it is a smart idea for Brian to hide in that room. You want to notify Brian that Alex has been in the room. The message is encoded with the same encoding matrix listed above, and the encoded message is the same as the one shown above as well.

To determine the decoding matrix, we must find the inverse of the encoding matrix, A^{-1} . For the purposes of this scenario, the formula method will be used and demonstrated through the following mathematical relationship:

$$A^{-1} = \frac{\text{adj } A}{\det A}$$

Our encoding matrix, A, is set up as such:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 8 & 3 \end{bmatrix}$$

Where each variable is equal to the number of the same position in the respective matrix. To solve find the 2 unknowns, we must manipulate the matrix as such:

$$\text{adj } A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\det A = ad - cb$$

Finding adj A, we see that the positions of the numbers and some of their values change. For this equation:

$$-b = -1 * 1 = -1$$

$$-c = -1 * 8 = -8$$

$$\text{adj } A = \begin{bmatrix} 3 & -1 \\ -8 & 3 \end{bmatrix}$$

And to find det A, we simply use the following equation below:

$$\det A = 3(3) - 1(-8)$$

$$\det A = 9 - 8$$

$$\det A = 1$$

So, to find A^{-1} :

$$A^{-1} = \frac{\text{adj } A}{\det A} = \begin{bmatrix} \frac{3}{1} & \frac{-1}{1} \\ \frac{-8}{1} & \frac{3}{1} \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 3 & -1 \\ -8 & 3 \end{bmatrix}$$

From here, the decoded message can be determined by the following formula

$$\text{Decoded Message} = A^{-1} * B$$

Each matrix must contain appropriate dimensions to be able to multiply. Here, the number of columns in A^{-1} must equal the number of rows in B. Since these both equal 2, the matrices can be multiplied as such:

$$\text{Decoded Message} = \begin{bmatrix} 3 & -1 \\ -8 & 3 \end{bmatrix} * \begin{bmatrix} 15 & 39 & 104 & 22 & 89 & 33 & 44 \\ 44 & 112 & 285 & 65 & 240 & 94 & 127 \end{bmatrix}$$

First row calculations:

$$3(15) - 1(44) = 1$$

$$3(39) - 1(112) = 5$$

$$3(104) - 1(285) = 27$$

$$3(22) - 1(65) = 1$$

$$3(89) - 1(240) = 27$$

$$3(33) - 1(94) = 5$$

$$3(44) - 1(127) = 5$$

Second row calculations:

$$-8(15) + 3(44) = 12$$

$$-8(39) + 3(112) = 24$$

$$-8(104) + 3(285) = 23$$

$$-8(22) + 3(65) = 19$$

$$-8(89) + 3(240) = 8$$

$$-8(33) + 3(94) = 18$$

$$-8(44) + 3(127) = 29$$

$$A^{-1} * B = \begin{bmatrix} 1 & 5 & 27 & 1 & 27 & 5 & 5 \\ 12 & 24 & 23 & 19 & 8 & 18 & 29 \end{bmatrix}$$

Using the cipher key above, Brian can lay out this matrix looking at the elements written per column:

1 12 5 24 27 23 1 19 27 8 5 18 5 29

Brian determines that the written message is:

ALEX WAS HERE!

Cipher Key:

A	B	C	D	E	F
1	2	3	4	5	6
G	H	I	J	K	L
7	8	9	10	11	12
M	N	O	P	Q	R
13	14	15	16	17	18
S	T	U	V	W	X
19	20	21	22	23	24
Y	Z		?	!	.
25	26	27	28	29	30

(3.) Scenario/Story for the 3 by 3 encoded secret message:

You are encouraging your son to work on his mannerisms his mother.

Additionally, your son is young and is struggling with the English language,

specifically spelling. However, he has a knack for math and memorizing numbers. You decide to try a method that you think may help him remember certain terms of gratitude while helping him memorize how to spell certain words. You develop the above Cipher Key for your son, and you want him to know the phrase:

THANK YOU MAMA.

Which is decoded as

20 8 1 14 11 27 25 15 21 27 13 1 13 1 30

and is shown in matrix form as such

$$\begin{bmatrix} 20 & 14 & 25 & 27 & 13 \\ 8 & 11 & 15 & 13 & 1 \\ 1 & 27 & 21 & 1 & 20 \end{bmatrix}$$

To find the decoded message, the decoding matrix A^{-1} is required and can be found using the formula method. In this scenario, the 3*3 encoding matrix being used is shown below.

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 1 & 1 \\ 2 & 2 & 1 \end{bmatrix}$$

This matrix is chosen because it is invertible and only contains positive integers. Additionally, the elements of the inverse of this matrix (the decoding matrix) have only integers as well. Now, using the encoding matrix A and the decoded message, we can find the encoded message. After checking the dimensions of each matrix and discovering the number of columns of the encoding matrix equal the number of rows of the decoded message, which is 3, the matrices are multiplied as such:

$$\text{Encoded Message} = B = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 1 & 1 \\ 2 & 2 & 1 \end{bmatrix} * \begin{bmatrix} 20 & 14 & 25 & 27 & 13 \\ 8 & 11 & 15 & 13 & 1 \\ 1 & 27 & 21 & 1 & 20 \end{bmatrix}$$

First row calculations:

$$1(20) + 2(8) + 2(1) = 20 + 16 + 2 = 38$$

$$1(14) + 2(11) + 2(27) = 14 + 22 + 54 = 90$$

$$1(25) + 2(15) + 2(21) = 25 + 30 + 42 = 97$$

$$1(27) + 2(13) + 2(1) = 27 + 26 + 2 = 55$$

$$1(13) + 2(1) + 2(20) = 13 + 2 + 40 = 55$$

Second row calculations:

$$1(20) + 1(8) + 1(1) = 20 + 8 + 1 = 29$$

$$1(14) + 1(11) + 1(27) = 14 + 11 + 27 = 52$$

$$1(25) + 1(15) + 1(21) = 25 + 15 + 21 = 61$$

$$1(27) + 1(13) + 1(1) = 27 + 13 + 1 = 41$$

$$1(13) + 1(1) + 1(20) = 13 + 1 + 20 = 34$$

Third row calculations:

$$2(20) + 2(8) + 1(1) = 40 + 16 + 1 = 57$$

$$2(14) + 2(11) + 1(27) = 28 + 22 + 27 = 77$$

$$2(25) + 2(15) + 1(21) = 50 + 30 + 21 = 101$$

$$2(27) + 2(13) + 1(1) = 54 + 26 + 1 = 81$$

$$2(13) + 2(1) + 1(20) = 26 + 2 + 20 = 48$$

$$B = \begin{bmatrix} 38 & 90 & 97 & 55 & 55 \\ 29 & 52 & 61 & 41 & 34 \\ 57 & 77 & 101 & 81 & 48 \end{bmatrix}$$

The encoded message can also be written out as

$$38 \ 29 \ 57 \ 90 \ 52 \ 77 \ 97 \ 61 \ 101 \ 55 \ 41 \ 81 \ 55 \ 34 \ 48$$

You write out the encoded message on your son's toy so he can remember this phrase.

(4.) Decoding the secret message encoded with 3 by 3 matrix:

We can now find the decoding matrix using the formula method.

$$A^{-1} = \frac{\text{adj } A}{\det A}$$

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 1 & 1 \\ 2 & 2 & 1 \end{bmatrix}$$

Each letter is equal to the number in its respective position. First, we find adj A:

$$\text{adj } A = \begin{bmatrix} ei - hf & hc - bi & bf - ec \\ gf - di & ai - gc & dc - af \\ dh - ge & gb - ah & ae - db \end{bmatrix}$$

$$ei - hf = 1(1) - 2(1) = 1 - 2 = -1$$

$$hc - bi = 2(2) - 2(1) = 4 - 2 = 2$$

$$bf - ec = 2(1) - 1(2) = 1 - 2 = 0$$

$$gf - di = 2(1) - 1(1) = 2 - 1 = 1$$

$$ai - gc = 1(1) - 2(2) = 1 - 4 = -3$$

$$dc - af = 1(2) - 1(1) = 2 - 1 = 1$$

$$dh - ge = 1(2) - 2(1) = 2 - 2 = 0$$

$$gb - ah = 2(2) - 1(2) = 4 - 2 = 2$$

$$ae - db = 1(1) - 1(2) = 1 - 2 = -1$$

$$\text{adj } A = \begin{bmatrix} -1 & 2 & 0 \\ 1 & -3 & 1 \\ 0 & 2 & -1 \end{bmatrix}$$

Then, we find $\det A$:

$$\det A = aei + bgf + cdh - ahf - bdi - cge$$

$$aei = 1 * 1 * 1 = 1$$

$$bgf = 2 * 2 * 1 = 4$$

$$cdh = 2 * 1 * 2 = 4$$

$$ahf = 1 * 2 * 1 = 2$$

$$bdi = 2 * 1 * 1 = 2$$

$$cge = 2 * 2 * 1 = 4$$

$$\det A = 1 + 4 + 4 - 2 - 2 - 4 = 1$$

Now, to find the decoding matrix:

$$A^{-1} = \frac{\text{adj } A}{\det A} = \begin{bmatrix} \frac{-1}{1} & \frac{2}{1} & \frac{0}{1} \\ \frac{1}{1} & \frac{-3}{1} & \frac{1}{1} \\ \frac{1}{1} & \frac{1}{1} & \frac{1}{1} \\ \frac{0}{1} & \frac{2}{1} & \frac{-1}{1} \\ \frac{1}{1} & \frac{1}{1} & \frac{1}{1} \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -1 & 2 & 0 \\ 1 & -3 & 1 \\ 0 & 2 & -1 \end{bmatrix}$$

From here, the decoded message can be determined by the following formula

$$\text{Decoded Message} = A^{-1} * B$$

Where each matrix must contain appropriate dimensions to be able to multiply.

Here, the number of columns in A^{-1} must equal the number of rows in B. Since these both equal 3, the matrices can be multiplied as such:

$$\text{Decoded Message} = \begin{bmatrix} -1 & 2 & 0 \\ 1 & -3 & 1 \\ 0 & 2 & -1 \end{bmatrix} * \begin{bmatrix} 38 & 90 & 97 & 55 & 55 \\ 29 & 52 & 61 & 41 & 34 \\ 57 & 77 & 101 & 81 & 48 \end{bmatrix}$$

First row calculations:

$$-1(38) + 2(29) + 0(57) = -38 + 58 = 20$$

$$-1(90) + 2(52) + 0(77) = -90 + 104 = 14$$

$$-1(97) + 2(61) + 0(101) = -97 + 122 = 25$$

$$-1(55) + 2(41) + 0(81) = -55 + 82 = 27$$

$$-1(55) + 2(34) + 0(48) = -55 + 68 = 13$$

Second row calculations:

$$1(38) - 3(29) + 1(57) = 38 - 87 + 57 = 8$$

$$1(90) - 3(52) + 1(77) = 90 - 156 + 77 = 11$$

$$1(97) - 3(61) + 1(101) = 97 - 183 + 101 = 15$$

$$1(55) - 3(41) + 1(81) = 55 - 123 + 81 = 13$$

$$1(55) - 3(34) + 1(48) = 55 - 102 + 48 = 1$$

Third row calculations:

$$0(38) + 2(29) - 1(57) = 0 + 58 - 57 = 1$$

$$0(90) + 2(52) - 1(77) = 0 + 104 - 77 = 27$$

$$0(97) + 2(61) - 1(101) = 0 + 122 - 101 = 21$$

$$0(55) + 2(41) - 1(81) = 0 + 82 - 81 = 1$$

$$0(55) + 2(34) - 1(48) = 0 + 68 - 48 = 20$$

$$\text{Decoded Message} = A^{-1} * B = \begin{bmatrix} 20 & 14 & 25 & 27 & 13 \\ 8 & 11 & 15 & 13 & 1 \\ 1 & 27 & 21 & 1 & 20 \end{bmatrix}$$

To determine the message, the matrix is laid out as follows:

20 8 1 14 11 27 25 15 21 27 13 1 13 1 30

and the message written is:

THANK YOU MAMA.

References:

Sullivan, M. (2019). Precalculus (11th Edition). Pearson.